

Lily Monte HW6

Lily Monte

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Code available at https://github.com/Lmonte695/plan372_homeworks

```
#setting working directory and loading in data  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2     3.5.1      v tibble     3.2.1  
## v lubridate  1.9.4      v tidyr      1.3.1  
## v purrr      1.0.2  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.4.3
```

```
library(stargazer)
```

```
##  
## Please cite as:  
##  
## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.  
## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```
rm(list = ls())
```

```
setwd("C:/Users/Lily/Documents/collegefiles/PLAN 372/plan372_hmks/hw6")
```

```
treedata = read_csv("TS3_Raw_tree_data.csv")
```

```
## Rows: 14487 Columns: 41  
## -- Column specification -----  
## Delimiter: ","  
## chr (15): Region, City, Source, Zone, Park/Street, SpCode, ScientificName, C...  
## dbl (25): DbaseID, cell, Age, DBH (cm), TreeHt (m), CrnBase, CrnHt (m), Cdia...
```

```
## num (1): TreeID
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Question 1

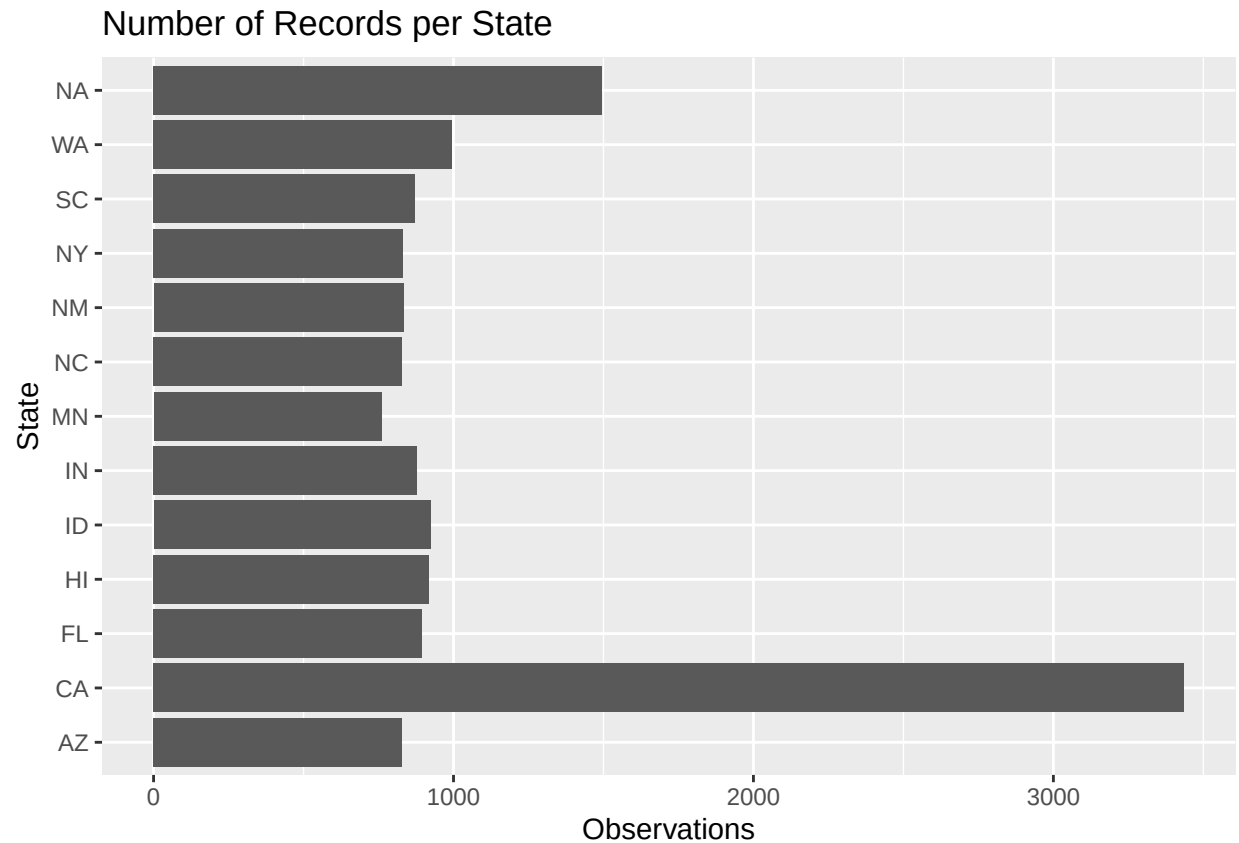
```
treedata = treedata %>%
  mutate(statecode = str_match(treedata$City, "^([:alnum:]+[:punct:]) ([:alnum:]+)"[,3]) %>%
    mutate(city = str_match(treedata$City, "^([:alnum:]+)([:punct:]) ([:alnum:]+)"[,2])

staterecords = treedata %>%
  group_by(statecode) %>%
  summarize(observations = n())

kable(staterecords, col.names = c("State", "Number of Observations"))
```

State	Number of Observations
AZ	827
CA	3435
FL	895
HI	918
ID	923
IN	877
MN	760
NC	828
NM	833
NY	831
SC	872
WA	994
NA	1494

```
ggplot() +
  geom_col(data = staterecords, aes(x = observations, y = statecode)) +
  labs(title = "Number of Records per State",
    y = "State",
    x = "Observations")
```



Question 2

```
treedatacleaned = treedata %>%
  filter(statecode == "NC" | statecode == "SC")
unique(treedatacleaned$city)
```

```
## [1] "Charleston" "Charlotte"
```

The two cities from which data were collected were Charleston, SC and Charlotte, NC

Question 3

```
treedatacleaned = treedatacleaned %>%
  mutate(genus = str_match(ScientificName, "^([:alnum:]+) ([:alnum:]+)")[,2])

canopy = treedatacleaned %>%
  group_by(genus) %>%
  summarize(avgdiam = mean(`AvgCdia (m)`)

canopy$genus[canopy$avgdiam == max(canopy$avgdiam)]
```

```
## [1] "Quercus"
```

```
max(canopy$avgdiam)
```

```
## [1] 13.62316
```

```
kable(canopy)
```

genus	avgdiam
Acer	11.451479
Betula	10.438462
Butia	4.690909
Carya	10.505556
Celtis	9.454054
Cornus	6.529412
Gleditsia	7.667568
Ilex	4.705634
Juniperus	6.009756
Lagerstroemia	6.259756
Liquidambar	9.980645
Magnolia	7.980488
Malus	6.631035
Pinus	8.655833
Platanus	12.014286
Prunus	9.268000
Pyrus	7.470000
Quercus	13.623163
Sabal	3.965000
Ulmus	9.962500

The genus with the largest crown diameter is the Quercus (oak), with an average crown diameter of 13.6 meters

Question 4

```
treedata$ScientificName = str_replace_all(treedata$ScientificName, " x ", "")

treedata = treedata %>%
  mutate(genus = str_match(ScientificName, "^([:alnum:]+) ([:alnum:]+)")[,2])

treedata = treedata %>%
  mutate(species = str_match(ScientificName, "^([:alnum:]+) ([:alnum:]+)")[,3])

genera = treedata %>%
  group_by(genus) %>%
  summarize(speciescount = n_distinct(species))

kable(genera, col.names = c("Genus", "Number of Extant Species"))
```

Genus	Number of Extant Species
Acacia	3
Acer	7
Aesculus	1
Betula	2
Brachychiton	1
Butia	1
Callistemon	1
Calocedrus	1
Calophyllum	1
Carpinus	1
Carya	1
Casuarina	1
Catalpa	1
Cedrus	1
Celtis	3
Ceratonia	1
Cercis	1
Chilopsis	1
Cinnamomum	1
Citharexylum	1
Cocos	1
Conocarpus	1
Cordia	1
Cornus	1
Crataegus	1
Cupaniopsis	1
Delonix	1
Elaeagnus	1
Elaeodendron	1
Eriobotrya	1
Eucalyptus	4
Fagus	1
Ficus	2
Filicium	1
Fraxinus	8
Ginkgo	1
Gleditsia	1
Gymnocladus	1
Ilex	2
Jacaranda	1
Juglans	1
Juniperus	1
Koelreuteria	2
Lagerstroemia	3
Liquidambar	1
Liriodendron	1
Magnolia	1
Malus	2
Melaleuca	1
Metrosideros	1
Morus	2
Olea	1

Genus	Number of Extant Species
Parkinsonia	2
Phoenix	2
Picea	1
Pinus	15
Pistacia	1
Pittosporum	1
Platanus	2
Platycladus	1
Podocarpus	1
Populus	5
Prosopis	1
Prunus	5
Pseudotsuga	1
Pyrus	3
Quercus	12
Rhus	1
Robinia	1
Sabal	1
Samanea	1
Schinus	2
Sequoia	1
Swietenia	1
Syagrus	1
Tabebuia	3
Tilia	3
Triadica	1
Tristaniopsis	1
Ulmus	4
Veitchia	1
Washingtonia	2
Zelkova	1
NA	1

Extra Credit

```
generaage = treedata %>%
  group_by(genus) %>%
  summarize(avgage = mean(Age))

summary(generaage$avgage)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    -1.00   20.12   24.29   26.23   30.76   69.96
```

```
generaage$avgage[generaage$genus == "Quercus"]
```

```
## [1] 28.99142      NA
```

Quercus, which has the largest average crown size, only has an average age slightly higher than the median. The large crown diameter of the Quercus genus is therefore likely not due to their age. It seems as if they simply have the largest crown diameter on average out of all tree Genera.

```
genera2 = treedatacleaned %>%
  group_by(genus) %>%
  summarize(across(c("Age", "AvgCdia (m)"), ~mean(.x, na.rm = T)))

kable(genera2)
```

genus	Age	AvgCdia (m)
Acer	24.47337	11.451479
Betula	18.97436	10.438462
Butia	20.63636	4.690909
Carya	28.97222	10.505556
Celtis	24.89189	9.454054
Cornus	17.39706	6.529412
Gleditsia	17.91892	7.667568
Ilex	22.76056	4.705634
Juniperus	26.48780	6.009756
Lagerstroemia	18.78049	6.259756
Liquidambar	25.04301	9.980645
Magnolia	29.09756	7.980488
Malus	12.10345	6.631035
Pinus	25.18333	8.655833
Platanus	27.73214	12.014286
Prunus	16.76000	9.268000
Pyrus	16.18571	7.470000
Quercus	35.55011	13.623163
Sabal	26.07500	3.965000
Ulmus	17.15625	9.962500

```
genera2$Age[genera2$genus == "Acer"]
```

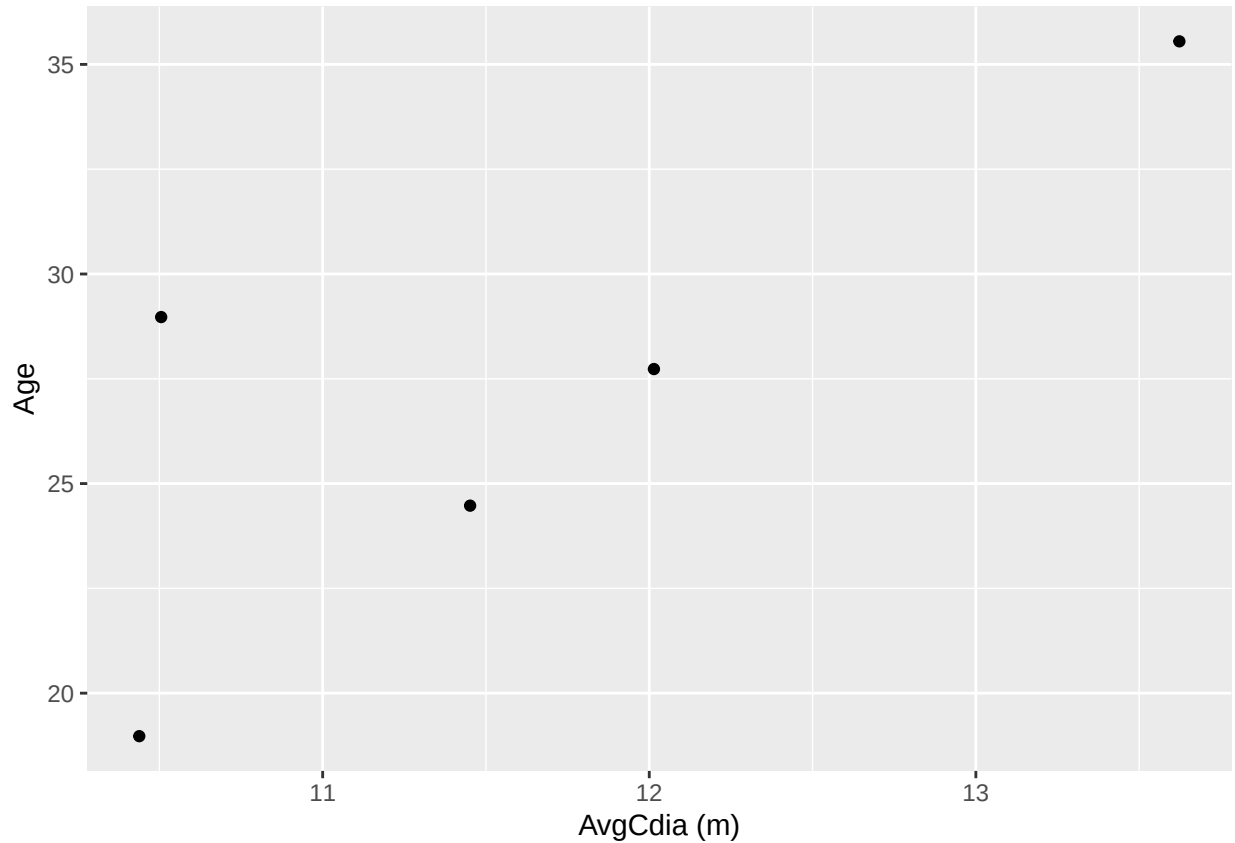
```
## [1] 24.47337
```

```
genera2$`AvgCdia (m)`[genera2$genus == "Acer"]
```

```
## [1] 11.45148
```

```
genera3 = genera2 %>%
  filter(`AvgCdia (m)` >= 10)

ggplot() +
  geom_point(data = genera3, aes(x = `AvgCdia (m)`, y = Age))
```



In North Carolina and south Carolina specifically, the genus *Quercus* is generally observed at older ages than other trees, with an average of 35.5 years old at the time of observation, as seen in the above table. So, they are likely not a tree with an especially large crown diameter compared to other genera. To determine the best quickly-growing, large tree for NC and SC, data was first filtered to only include trees with a diameter of 10 or greater. Plotting this, with the exception of one genus with a high age but comparatively low average crown diameter, the relationship between age and crown diameter is almost perfectly linear. Thus, *Acer*, *Betula*, *Platanus*, and *Quercus* are all feasible genera for the recommendation, as they all produce relatively comparative ratios of crown diameter to age.